

PCA - Principle Component Analysis

Unsupervised statistical technique used to examine the interrelations among a set of variables in order to identify the underlying structure of those variables.

Also known as general factor analysis.

- Regression Analysis $\rightarrow$ line of best fit to a data set
- Factor Analysis $\rightarrow$ several orthogonal lines of best fit to the data set.
 $\downarrow$
 right angles
 $\downarrow$

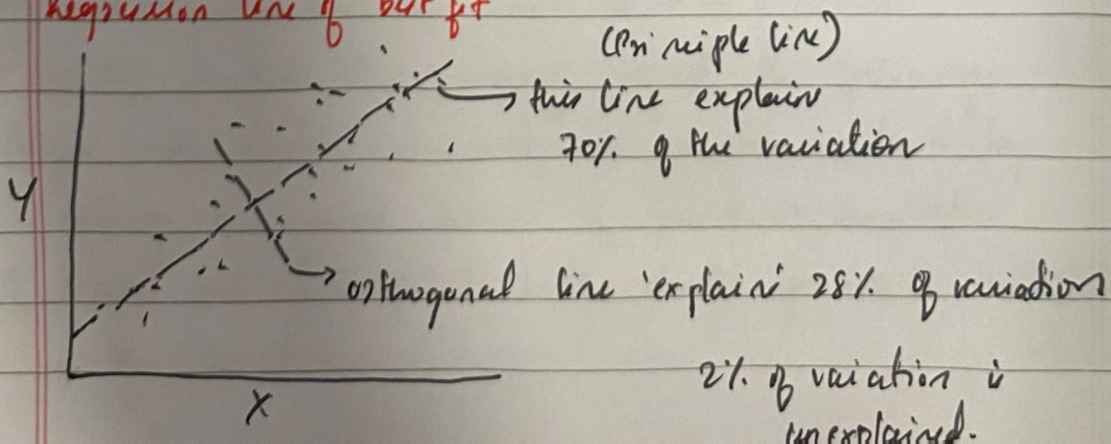
These lines are perpendicular to each other in n -dimensional space.

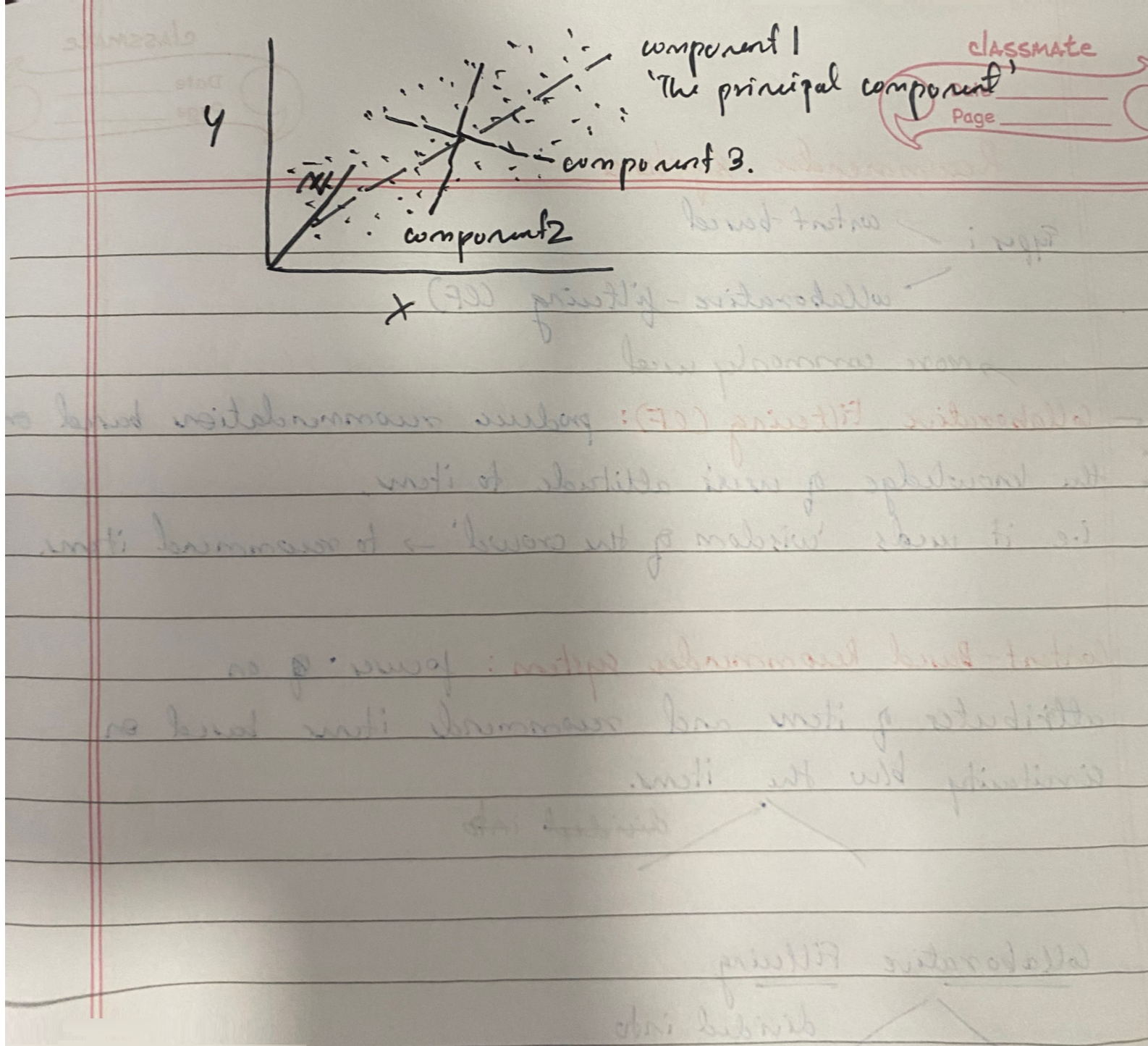
- **N-Dimensional Space:** variable sample space.

of dimensions = # of variables.

In a dataset with 4 variables, the sample space is 4-dimensional.

Regression line of best fit





Recommender Systems

Types: / content-based

collaborative-filtering (CF)

→ more commonly used

Has
ability —
to learn
features
on its
own

Collaborative Filtering (CF): produces recommendation based on the knowledge of user's attitude to items.
i.e. it uses 'wisdom of the crowd' → to recommend items.

Content-Based Recommender system: focuses on attributes of items and recommends items based on similarity b/w the items.

divided into

Collaborative Filtering

divided into

Memory-based

(cosine similarity)

Model-based

(singular value decomposition)